

DBMS – PRODUCT AND CATEGORY MANAGEMENT

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	INT	Primary Key
Category_Name	VARCHAR(100)	NOT NULL, UNIQUE
Description	VARCHAR(200)	—

1.2 Brand Table

Attribute	Data Type	Constraint
Brand_ID	INT	Primary Key
Brand_Name	VARCHAR(100)	NOT NULL, UNIQUE
Country	VARCHAR(50)	—
Description	VARCHAR(200)	—

1.3 Product Table

Attribute	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	VARCHAR(100)	NOT NULL
Brand_ID	INT	Foreign Key
Category_ID	INT	Foreign Key
Product_Size	VARCHAR(20)	—
Color	VARCHAR(30)	—
Price	NUMBER(10,2)	NOT NULL
Stock	INT	NOT NULL
Description	VARCHAR(200)	—

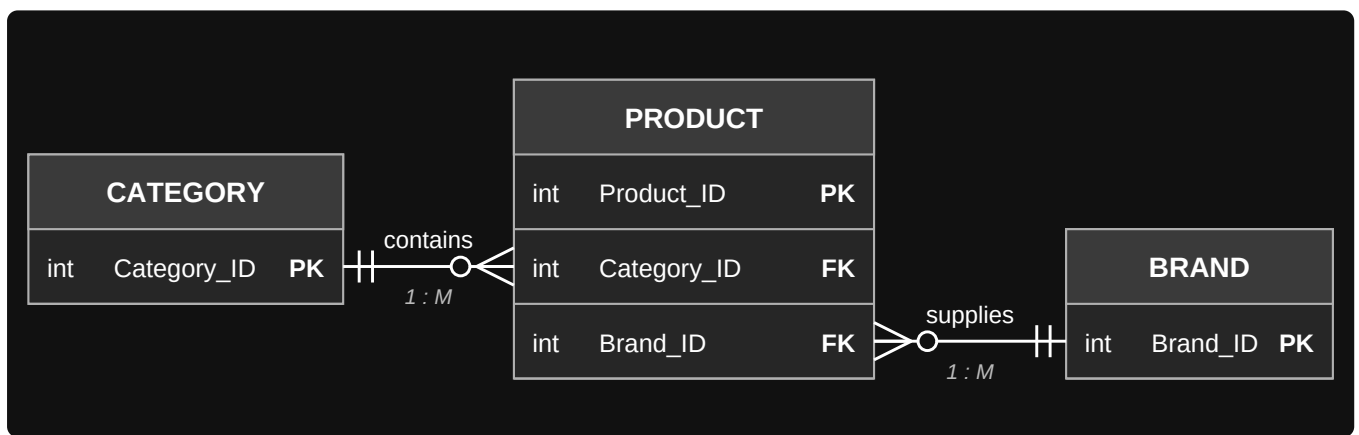
2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: **One-to-Many (1:M)**

Brand → Product

- Brand.Brand_ID → Primary Key
- Product.Brand_ID → Foreign Key referencing Brand.Brand_ID
- Relationship: **One-to-Many (1:M)**



3. Table Creation

3.1 Create Category Table

```

SQL> CREATE TABLE Category (
2     Category_ID INT PRIMARY KEY,
3     Category_Name VARCHAR(100) NOT NULL UNIQUE,
4     Description VARCHAR(200)
5 );

```

Table created.

3.2 Create Brand Table

```

SQL> CREATE TABLE Brand (
2     Brand_ID INT PRIMARY KEY,
3     Brand_Name VARCHAR(100) NOT NULL UNIQUE,
4     Country VARCHAR(50),
5     Description VARCHAR(200)
6 );

```

Table created.

3.3 Create Product Table

```

SQL> CREATE TABLE Product (
2     Product_ID INT PRIMARY KEY,
3     Product_Name VARCHAR(100) NOT NULL,
4     Brand_ID INT NOT NULL,
5     Category_ID INT NOT NULL,
6     Product_Size VARCHAR(20),
7     Color VARCHAR(30),
8     Price NUMBER(10,2) NOT NULL,
9     Stock INT NOT NULL,
10    Description VARCHAR(200),
11    FOREIGN KEY (Brand_ID) REFERENCES Brand(Brand_ID),
12    FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
13 );

```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category
  2  VALUES (1, 'Men Clothing', 'Casual and formal wear for men');

1 row created.

SQL> INSERT INTO Category
  2  VALUES (2, 'Women Clothing', 'Ethnic and western wear for women');

1 row created.

SQL> INSERT INTO Category
  2  VALUES (3, 'Footwear', 'Shoes, sandals and sports footwear');

1 row created.

SQL> INSERT INTO Category
  2  VALUES (4, 'Accessories', 'Wallets, belts, watches and bags');

1 row created.

SQL> INSERT INTO Category
  2  VALUES (5, 'Kids Wear', 'Clothing for boys and girls');

1 row created.
```

4.2 Brand Records

```
SQL> INSERT INTO Brand
  2  VALUES (1, 'Roadster', 'India', 'Casual fashion brand');

1 row created.

SQL> INSERT INTO Brand
  2  VALUES (2, 'Biba', 'India', 'Ethnic wear brand');

1 row created.

SQL> INSERT INTO Brand
  2  VALUES (3, 'Nike', 'USA', 'Sports and footwear brand');

1 row created.

SQL> INSERT INTO Brand
  2  VALUES (4, 'Puma', 'Germany', 'Sportswear and accessories brand');

1 row created.

SQL> INSERT INTO Brand
  2  VALUES (5, 'HRX', 'India', 'Activewear and kids fashion');

1 row created.
```

4.3 Product Records

```
SQL> INSERT INTO Product
  2 VALUES (101, 'Slim Fit Cotton Shirt', 1, 1, 'M', 'Blue', 899.00, 50, 'Men slim fit casual shirt');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (102, 'Printed Cotton Kurta', 2, 2, 'L', 'Pink', 1299.00, 40, 'Women printed ethnic kurta');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (103, 'Running Shoes', 3, 3, '9', 'Black', 3499.00, 25, 'Lightweight running shoes');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (104, 'Leather Wallet', 4, 4, 'Free', 'Brown', 599.00, 15, 'Genuine leather wallet');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (105, 'Kids Cotton T-Shirt', 5, 5, '6-7Y', 'Yellow', 399.00, 60, 'Soft cotton t-shirt for kids');

1 row created.

SQL> COMMIT;

Commit complete.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SET LINESIZE 200
SQL> SET PAGESIZE 50
SQL> COLUMN Description FORMAT A35
SQL> COLUMN Product_Name FORMAT A25

SQL> SELECT * FROM Category;

CATEGORY_ID CATEGORY_NAME DESCRIPTION
-----
          1 Men Clothing  Casual and formal wear for men
          2 Women Clothing Ethnic and western wear for women
          3 Footwear      Shoes, sandals and sports footwear
          4 Accessories   Wallets, belts, watches and bags
          5 Kids Wear     Clothing for boys and girls

5 rows selected.
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;

PRODUCT_ID PRODUCT_NAME          BRAND_ID CATEGORY_ID PRODUCT_SIZE COLOR  PRICE STOCK DESCRIPTION
-----
        101 Slim Fit Cotton Shirt          1          1 M      Blue   899   50 Men slim fit casual shirt
        102 Printed Cotton Kurta          2          2 L      Pink  1299   40 Women printed ethnic kurta
        103 Running Shoes                3          3 9      Black 3499   25 Lightweight running shoes
        104 Leather Wallet                4          4 Free   Brown  599   15 Genuine leather wallet
        105 Kids Cotton T-Shirt          5          5 6-7Y   Yellow 399   60 Soft cotton t-shirt for kids

5 rows selected.
```

6. Product Update

Update the price and stock of Slim Fit Cotton Shirt.

```
SQL> UPDATE Product
  2  SET Price = 799.00,
  3      Stock = 45
  4  WHERE Product_ID = 101;

1 row updated.
```

7. Product Deletion

Delete the product Kids Cotton T-Shirt.

```
SQL> DELETE FROM Product
  2  WHERE Product_ID = 105;

1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category.

```
SQL> SELECT
  2  c.Category_Name,
  3  p.Product_Name,
  4  p.Price,
  5  p.Stock
  6  FROM Category c
  7  JOIN Product p
  8  ON c.Category_ID = p.Category_ID
  9  ORDER BY c.Category_Name;

CATEGORY_NAME  PRODUCT_NAME          PRICE STOCK
-----
Accessories    Leather Wallet         599    15
Footwear       Running Shoes         3499   25
Men Clothing   Slim Fit Cotton Shirt   799    45
Women Clothing Printed Cotton Kurta  1299   40

4 rows selected.
```

8.2 Products in Men Clothing Category

```
SQL> SELECT
  2  p.Product_ID,
  3  p.Product_Name,
  4  p.Price,
  5  p.Stock
  6  FROM Product p
  7  JOIN Category c
  8  ON p.Category_ID = c.Category_ID
  9  WHERE c.Category_Name = 'Men Clothing';

PRODUCT_ID  PRODUCT_NAME          PRICE STOCK
-----
101 Slim Fit Cotton Shirt   799    45
```

8.3 Product Count by Category

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Total_Products
  4 FROM Category c
  5 LEFT JOIN Product p
  6 ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name
  8 ORDER BY c.Category_Name;
```

CATEGORY_NAME	TOTAL_PRODUCTS
Accessories	1
Footwear	1
Kids Wear	0
Men Clothing	1
Women Clothing	1

5 rows selected.